

Technical Whitepaper: The HexaHelix B-N-C Architecture

Resonant Magnetohydrodynamics and Direct Energy Conversion in Compact High-Field Stellarators

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Abstract

The HexaHelix architecture proposes solutions to two major challenges in magnetic confinement fusion: the inefficiency of conventional thermal energy conversion and the manufacturing complexity associated with non-planar high-temperature superconductor (HTS) coils. By implementing a six-fold interlaced helical geometry, optimized through AI-augmented simulations, and employing a Boron-Nitrogen-Carbon (B-N-C) metamaterial, the system can stabilize magnetic fields surpassing 30 tesla within compact volumes. A new Direct Energy Conversion (DEC) module enables resonant inductive extraction of kinetic energy from charged particles, replacing the traditional Carnot-limited thermal cycle. Recent quantum studies on engineered reservoirs and non-local energy transfer support the feasibility of low-loss transmission in the DEC approach. Early simulations suggest efficiencies up to 70%, along with notable reductions in system size and cost, making modular and commercially feasible fusion reactors possible.

1. Introduction

Magnetic confinement fusion has advanced considerably, with devices like Wendelstein 7-X achieving notable quasi-isodynamic (QI) performance. Nevertheless, commercial scalability is limited by significant thermal losses and the manufacturing difficulties posed by non-planar HTS coils subjected to extreme mechanical stress. The HexaHelix method, developed through the human-AI collaborative Twin-Code methodology, introduces a topology that integrates geometric resilience, innovative materials, and direct energy extraction. This architecture is designed to transition stellarators from research instruments to efficient, modular solid-state energy generators.

2. System Architecture: 6-Fold Interlaced Helix

The main magnet configuration consists of six identical helical modules, interlaced and arranged in a hexagonal symmetry:

- Golden Ratio Twist ($\varphi \approx 0.618$): Enhances magnetic shear and magnetohydrodynamic (MHD) self-stabilization, while minimizing Lorentz force shear stresses.

- **Parallel Manufacturing:** Utilizing six identical modules enables parallel production lines, greatly reducing lead times and complexity compared to unique, large-scale assemblies.
- **Redundancy by Design:** The interlaced arrangement provides fault tolerance, with simulations demonstrating stable confinement even if some segments fail, thanks to compensatory field superposition.
- **Scaling Law:** At magnetic field strengths exceeding 30 T, the required plasma volume for achieving $Q > 1$ decreases significantly, allowing for a more compact reactor footprint.

3. Materials Science: B-N-C Strain Engineering & ISD-Platform

The HexaHelix coils incorporate a B-N-C metamaterial architecture, produced through scalable Chemical Vapor Deposition (CVD):

- **ISD-MgO Advantage (No-Plasma):** The design is based on an ISD-MgO (Inclined Substrate Deposition) template, which enables biaxial texturing without destructive ion bombardment typical in IBAD processes, thereby maintaining the atomic purity of the sensitive sp²-hybrid lattice.
- **Internal Pressure Hack:** The synergistic co-doping of Carbon lattices with Boron-11 and Nitrogen generates localized electric fields that simulate multi-gigapascal external pressure. This atomic-scale strain engineering increases the critical temperature and magnetic field stability, eliminating the need for extensive steel containment.
- **Resilient Supply Chain:** Unlike conventional REBCO tapes, the B-N-C approach uses abundant, low-cost industrial precursors such as hydrocarbons, enriched boric acid, and melamine, resulting in reduced material capital expenditures and a more resilient supply chain.
- **Neutron Resilience:** The exclusive use of Boron-11 ($\sigma \approx 0.005$ barns) minimizes activation and radiation damage compared to natural boron.
- **Thermal Management:** The coils feature an intrinsic 3D-porous architecture (~100 nm pores), which allows efficient coolant flow and achieves ten times greater heat rejection compared to solid coils.

4. Energy Extraction: Inductive Direct Energy Conversion (DEC)

Traditional fusion reactors depend on neutron capture and thermal cycles that are limited by Carnot efficiency (~40%). The HexaHelix design introduces resonant inductive coupling for direct energy conversion, acting as an electromagnetic damper for plasma oscillations.

- **Dual Circuit Design:** Two separate coil subsystems, both leveraging B-N-C material but with distinct structures, are used:
- **Magnet Circuit (DC – Steady-State Confinement):** Features a dense, compact B-N-C layer on an ISD-MgO substrate to generate a stable magnetic field (>30 T) for plasma confinement. This layer is solid-state, with no pores, to maximize density and stability, supported by the internal pressure hack for simulating gigapascal pressures. DC operation demands stability with constant current.

- **Generator Circuit (AC – Energy Extraction):** Utilizes a nano-porous B-N-C architecture (~100 nm pores) as a liner facing the plasma, also on the ISD-MgO substrate. This layer acts as a resonant receiver for DEC, using plasma oscillations for inductive energy conversion. The pores optimize flux penetration and coupling. AC operation requires resonance and damping to extract energy and stabilize plasma.

Integration & Benefits: Both circuits share the ISD-MgO substrate at room temperature, minimizing mismatches such as thermal stresses and cracks. The generator is positioned as the inner layer (plasma-facing), while the magnet forms the outer stable core, preventing interference between DC and AC operations. The generator damps plasma oscillations, reducing quench risks for the magnet. Projected efficiency exceeds 70%, with compact scaling (e.g., Power-Box dimensions of 2x2x4 m).

Resonance Condition: The natural frequency of the B-N-C lattice is tuned to match plasma oscillation frequency, maximizing mutual inductance through the porous structure.

Direct Efficiency: Theoretical models indicate up to 70% efficiency by directly converting kinetic energy to induced voltage, eliminating the need for steam turbines.

5. Economic Scaling: The 150 MW Pilot Module

The 150 MW industrial pilot module is designed as the smallest commercially viable unit, optimized for factory production and standard logistics:

- **Factory-Built & Modular:** Modules (~5m x 6m) are suitable for parallel manufacturing and on-site torus integration, simplifying transport logistics.
- **CAPEX Reduction:** By removing the need for complex steam infrastructure and using standard industrial CVD equipment, significant capital expense savings are achieved.
- **LCOE Target:** Miniaturization and direct conversion enable an optimized Levelized Cost of Electricity (LCOE) for decentralized, high-availability energy markets.

6. Conclusion

The HexaHelix B-N-C architecture reimagines fusion energy from "brute force" engineering to "smart materials" engineering. By integrating AI-optimized geometry, leveraging the manufacturing advantages of ISD-MgO, and implementing resonant energy extraction, this approach offers exponential potential for modular, economically viable fusion energy.

Open Repository: github.com/Haegar1601/hexa-helix-stellarator

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This whitepaper is generated through Twin-Code Strategy (Human Intuition + AI Logic). Feedback and collaboration welcome.